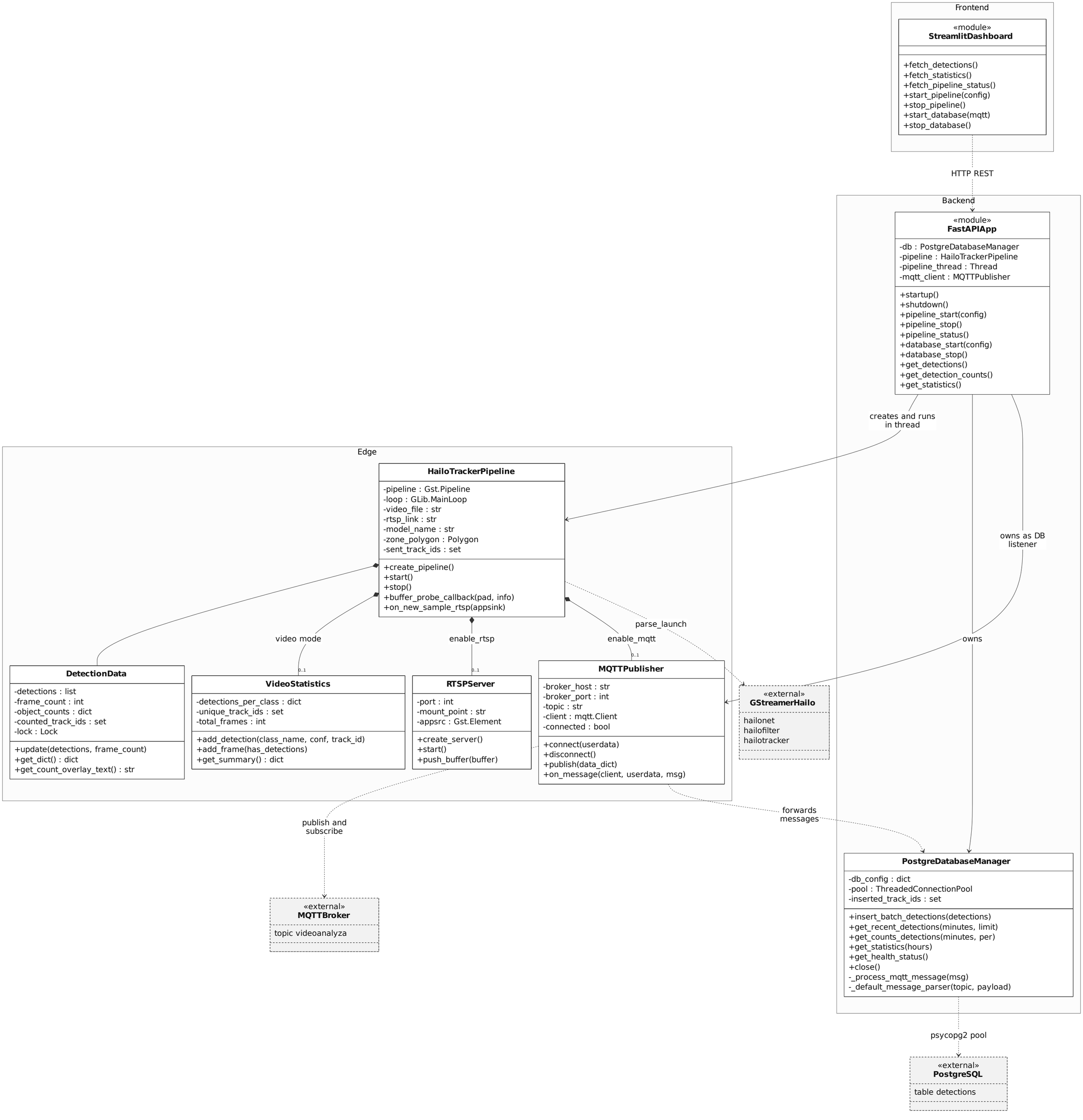

Frontend
«module»
StreamlitDashboard
+fetch_detections()
+fetch_statistics()
+fetch_pipeline_status()
+start_pipeline(config)
+stop_pipeline()
+start_database(mqtt)
+stop_database()
HTTP REST
Backend
«module»
FastAPIApp
-db : PostgreDatabaseManager
-pipeline : HailoTrackerPipeline
-pipeline_thread : Thread
-mqtt_client : MQTTPublisher
+startup()
+shutdown()
+pipeline_start(config)
+pipeline_stop()
+pipeline_status()
+database_start(config)
+database_stop()
+get_detections()
+get_detection_counts()
+get_statistics()
creates and runs in thread
owns as DB listener
Edge
HailoTrackerPipeline
-pipeline : Gst.Pipeline
-loop : GLib.MainLoop
-video_file : str
-rtsp_link : str
-model_name : str
-zone_polygon : Polygon
-sent_track_ids : set
+create_pipeline()
+start()
+stop()
+buffer_probe_callback(pad, info)
+on_new_sample_rtsp(appsink)
video mode
enable_rtsp
enable_mqtt
parse_launch
owns
0..1
DetectionData
-detections : list
-frame_count : int
-object_counts : dict
-counted_track_ids : set
-lock : Lock
+update(detections, frame_count)
+get_dict() : dict
+get_count_overlay_text() : str
VideoStatistics
-detections_per_class : dict
-unique_track_ids : set
-total_frames : int
+add_detection(class_name, conf, track_id)
+add_frame(has_detections)
+get_summary() : dict
RTSPServer
-port : int
-mount_point : str
-appsrc : Gst.Element
+create_server()
+start()
+push_buffer(buffer)
MQTTPublisher
-broker_host : str
-broker_port : int
-topic : str
-client : mqtt.Client
-connected : bool
+connect(userdata)
+disconnect()
+publish(data_dict)
+on_message(client, userdata, msg)
«external»
GStreamerHailo
hailonet
hailofilter
hailotracker
publish and subscribe
forwards messages
«external»
MQTTBroker
topic videoanalyza
PostgreDatabaseManager
-db_config : dict
-pool : ThreadedConnectionPool
-inserted_track_ids : set
+insert_batch_detections(detections)
+get_recent_detections(minutes, limit)
+get_counts_detections(minutes, per)
+get_statistics(hours)
+get_health_status()
+close()
-_process_mqtt_message(msg)
-_default_message_parser(topic, payload)
psycopg2 pool
«external»
PostgreSQL
table detections